

LANCASTER, TX, USA

WMO: 722143

Lat: 32.579N

Lon: 96.719W

Elev: 501

StdP: 14.43

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	27.4	10.2	9.4	30.2	14.9	11.9	35.1	26.6	47.5	24.3	48.7	6.7	0	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	100.1	73.8	98.7	73.7	96.4	73.9	78.3	89.0	77.5	89.3	76.8	88.5	7.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	136.2	81.7	74.6	132.1	81.3	73.4	126.5	80.8	42.1	88.7	41.2	89.5	40.3	88.8	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	19.1	17.1	16.6	103.6	4.0	3.0	13.7	105.7	11.3	107.5	9.1	109.1	6.2	111.3
			15.2	79.6	3.8	1.4	12.5	80.6	10.3	81.4	8.1	82.2	5.4	83.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.7	45.7	49.5	57.4	64.5	72.7	80.9	83.7	84.2	78.1	66.9	55.8	47.6
	DBStd	15.72	9.51	10.44	9.24	7.60	6.37	4.23	4.06	4.66	5.71	8.31	9.22	9.23
	HDD50	573	197	128	38	3	0	0	0	0	0	3	45	159
	HDD65	2340	600	442	263	99	16	0	0	0	1	77	299	543
	CDD50	6289	63	113	268	438	702	928	1044	1059	843	527	219	85
	CDD65	2582	2	7	29	84	253	478	579	594	393	136	23	4
	CDH74	29116	20	45	177	636	2281	5414	7221	7556	4271	1325	144	26
	CDH80	14480	2	3	16	132	812	2707	3975	4270	2086	453	22	2
Wind	WSAvg	6.8	7.4	8.1	8.5	8.5	7.2	6.5	5.5	5.1	5.0	6.0	7.0	6.9
Precipitation	PrecAvg	38.4	2.7	2.6	3.5	3.7	4.8	3.6	2.0	2.2	2.8	4.3	3.1	3.0
	PrecMax	66.3	6.3	7.4	7.0	7.3	15.2	10.7	4.9	5.1	9.3	11.7	10.2	8.7
	PrecMin	20.9	0.2	0.2	0.1	1.2	0.6	0.7	0.0	0.0	0.5	0.3	0.5	0.2
	PrecStd	10.0	1.6	1.8	1.9	1.8	2.8	2.6	1.5	1.4	2.2	3.0	2.3	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	79.2	82.1	88.1	93.2	99.3	102.4	104.1	99.2	91.9	82.4	77.3
		MCWB	61.3	61.8	64.6	70.5	73.1	72.9	74.0	73.5	72.5	71.2	67.0	63.5
	2%	DB	71.7	74.5	78.8	83.1	90.4	96.9	99.7	101.0	96.2	88.6	78.1	72.4
		MCWB	57.7	62.0	64.9	67.9	72.6	73.1	74.0	73.9	73.2	70.3	65.0	61.9
	5%	DB	66.2	70.8	75.4	80.8	88.0	94.6	97.7	99.3	93.4	85.4	74.0	68.1
		MCWB	56.8	60.6	63.0	67.0	72.0	74.1	74.1	73.8	73.2	68.8	63.6	59.8
	10%	DB	62.9	67.5	72.4	77.5	84.5	92.0	95.3	96.8	90.9	81.6	71.4	63.9
		MCWB	55.2	59.7	62.3	65.7	70.9	74.1	74.6	74.2	72.1	66.7	63.0	56.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	68.2	69.3	73.4	76.9	79.8	79.8	78.5	78.2	75.9	70.6	67.9
		MCDB	71.7	74.3	75.9	82.3	86.5	88.8	89.0	89.8	87.6	83.0	76.6	73.5
	2%	WB	62.7	66.0	67.5	71.4	75.4	78.1	78.3	77.6	76.7	73.9	68.6	65.1
		MCDB	66.3	71.1	74.3	79.3	85.2	88.2	89.3	91.0	87.5	82.8	73.8	70.1
	5%	WB	59.8	63.6	66.2	70.0	74.2	77.1	77.4	76.9	75.6	72.3	66.5	62.0
		MCDB	64.9	68.1	72.7	76.7	83.6	87.7	89.3	90.4	86.4	80.7	72.4	66.2
	10%	WB	55.1	60.6	64.4	68.5	73.0	76.0	76.6	76.1	74.6	70.7	64.0	58.3
		MCDB	61.8	66.0	70.4	75.1	81.8	86.8	88.4	89.4	85.2	77.9	70.0	63.2
Mean Daily Temperature Range	5% DB	MDBR	20.5	20.1	20.9	21.2	19.9	20.2	21.3	21.8	22.0	22.7	21.3	20.0
		MCDBR	27.1	25.6	23.1	22.6	22.6	23.3	23.9	25.0	24.5	25.2	23.8	25.9
		MCWBR	17.5	16.1	12.9	11.4	8.4	6.4	6.0	6.3	7.7	10.5	13.8	17.3
	5% WB	MCDBR	22.5	20.5	19.1	18.1	18.7	19.0	20.4	21.3	20.3	19.9	19.6	21.2
		MCWBR	18.3	15.1	12.4	11.2	8.4	7.2	6.7	6.2	7.0	10.1	14.5	17.5
Clear-Sky Solar Irradiance	taub	0.317	0.332	0.359	0.399	0.428	0.441	0.444	0.445	0.403	0.360	0.341	0.321	
	taud	2.499	2.462	2.405	2.314	2.291	2.288	2.319	2.297	2.368	2.473	2.468	2.489	
	Ebn at Noon	287	293	292	283	274	269	268	266	274	279	275	279	
	Edh at Noon	27	31	36	41	42	42	41	41	37	31	28	26	
All-Sky Solar Radiation	RadAvg	908	1051	1401	1706	1840	2070	2116	1946	1637	1287	963	801	
	RadStd	88	168	113	126	158	192	146	129	137	191	122	104	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page